

Vyāyāma

An AI exercise coach that runs entirely on your phone.

Good form is the line between progress and injury — but live feedback usually means a personal trainer, a wearable, or a cloud vision API. Train at home and you get none of it; most “AI fitness” apps stream your camera to someone else’s server.

Vyāyāma puts the coach on-device — real-time reps + form on the Snapdragon NPU. No trainer, no wearable, no cloud. Your camera feed never leaves the phone.

Hack4SoC 3.0 · On-Device / Edge AI · Qualcomm

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व्यायाम (vyāyāma) — Sanskrit for “exercise.” · 0 bytes leave the device.

On-device pose intelligence, end to end

THE STACK

- **Android · Java · Camera2** YUV preview, zero-copy frames
- **Qualcomm SNPE → Hexagon NPU** INT8 .dlc models, GPU/CPU fallback
- **YOLO-NAS + HRNet** person box → 17 COCO keypoints
- **VyāyamaCoach engine** pure-Java, zero-alloc per frame
- **Offline storage** SharedPreferences · AlarmManager reminders
- **No backend · no network** the app requests no INTERNET permission

WHAT IT DOES

- **Sees your skeleton in real time** recognizes the move (7 exercises)
- **Counts reps + scores form 0–100** live cues: “lower.. now drive up!”
- **Manual mode** pin one exercise so it can’t misread
- **Offline profiles** personal bests, streaks, daily reminders
- **Coach Vision overlay** shows exactly what the engine senses

DETAIL Zero heap allocation per frame — ring buffers reused, so the GC never stutters mid-rep.

DETAIL One-Euro filter smooths jitter without adding lag; NPU by default, fallback in one tap.

BUILT TODAY 7 exercises auto-recognized · live NPU pipeline (GPU/CPU fallback) · manual mode · offline profiles, streaks & reminders · 105 self-tests green.

Camera to coaching — every stage on-device

ON-DEVICE · NO INTERNET PERMISSION

CAPTURE

Camera2

YUV · 30 fps

active-athlete lock

upright rotate

NPU INFERENCE

SNPE · Hexagon

YOLO-NAS · person box

HRNet · 17 keypoints

INT8 .dlc

SIGNAL → FEATURES

robust prep

One-Euro smoothing

teleport reject + gap-hold

13 biomech features

COACH ENGINE

pure-Java · 0-alloc

sticky recognizer

rep FSM ±partial/fast

adaptive ROM

form 0–100 + cue

🔄 profiles + calibration personalize each session

EXPERIENCE

the app

Live HUD

Coach Vision

Profiles · PB · streak

Reminders → notifications

EFFICIENCY INT8 on the Hexagon NPU · zero heap-alloc per frame · GPU/CPU fallback in one tap · the engine is validated by a 105-assertion offline test harness.

A coach for everyone — private by design

SOCIAL RELEVANCE

- **Access & equity** trainer-grade form for the millions with no gym, coach or PT
- **Privacy & dignity** camera never leaves the phone — safe for women, minors, home & clinical use
- **Rehab & ageing** guided reps for physiotherapy and elderly home workouts between clinic visits
- **Reach** fully offline — tier-2/3, rural, low-connectivity; budget phone to flagship

POTENTIAL IMPACT

- **Fewer injuries, real adherence** live form turns risky reps into safe ones, so people keep going
- **Remote physiotherapy** track recovery at home and ease the load on clinics
- **Privacy-first health AI** a template: genuinely useful AI that collects zero data
- **Scales at ~zero cost** no servers, no cloud bill, no data-centre energy — millions, on-device

THE BIG PICTURE On-device AI makes form-correct exercise a free, private, offline utility — not a subscription that watches you.

Beyond a rep counter — the details that add up

ROBUST RECOGNITION

- Sticky self-correcting lock — flicker can't switch it
- Positive-evidence gates: a sit-up is never a curl
- Catches rep #1 even if recognized mid-move
- Manual mode: pin one move, zero misreads

ADAPTIVE COUNTING

- Two-threshold FSM rejects jitter + partials
- Learns your range — low-ROM reps count
- Hip-drop signal counts foreshortened squats
- Fast reps at low frame-rate still register

REAL-TIME COACHING

- Per-rep form score 0–100
- One cue, worst issue first (depth, sag, swing...)
- Live phase cues: "lower... now drive up!"
- PB rewards · streaks · daily reminders

UX TOUCHES

- Animated splash: running athlete + wordmark
- Volt HUD: big counter, colour-coded cues
- Stat strip (streak / today's best) over camera
- Tap to rotate · engine switch (NPU/GPU/CPU)

EFFICIENT & PRIVATE

- Zero heap allocation per frame — no GC jank
- INT8 on the Hexagon NPU — battery-light
- GPU / CPU fallback in one tap
- No INTERNET permission — provably offline

ENGINEERING RIGOR

- 105-assertion pure-Java harness (offline, ms)
- Every module mocked — runs in any subset
- Hardened via multi-agent adversarial review
- Coach Vision = built-in transparency

Your phone already has the hardware — Vyāyāma turns it into a coach: private, offline, real-time.

NEXT more movements + rehab · voice coaching (eyes-off) · on-device personalization · optional HR fusion — still offline